

Context-oriented Synthesis of Salt Domes in Labeled Seismic Images

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ABSTRACT Advanced remote sensing and seismic image analysis and interpretation methods based on training robust machine learning models, which rely on large annotated datasets. However, obtaining such large annotated datasets is challenging in applications like offshore oil exploration, where expert annotation is costly and time-consuming. This work proposes a context-oriented data augmentation scheme for generating synthetic seismic images of salt dome structures with pixel-level mask labels. The proposed approach uses Variational Autoencoders (VAE) to learn and generate realistic salt body geometries, that are combined with a context-aware non-parametric texture synthesis method, to create seismic textures across three distinct zones: salt bodies, conventional rocks, and their boundaries. Using the TGS Salt Identification Challenge dataset as training data, several generated synthetic samples were evaluated by three geoscientists and compared with a state-of-the-art seismic image synthesis method. ~~The qualitative evaluation shows that the~~ **A blind discrimination experiment — where real and synthetic images are virtually indistinguishable from the real ones, with the experts F1-scores differing by less than 2%.** ~~presented in randomized, interleaved order — is currently being conducted to rigorously assess the perceptual realism of the synthesized images.~~ **[TODO: Update abstract with blind experiment results when available.]** Furthermore, quantitative comparisons demonstrate substantial improvements over the baseline, achieving significantly lower pixel-wise error, higher structural similarity, and better texture fidelity. These results suggest the effectiveness of the proposed context-oriented approach for generating high-quality synthetic seismic training data, which is crucial for seismic image analysis and interpretation.

INDEX TERMS Data augmentation, deep learning, image segmentation, seismic images, seismic interpretation, texture synthesis, and variational autoencoders.

I. INTRODUCTION

A Relevant task in seismic imaging analysis and interpretation is to distinguish accurately saline bodies from other sediments. However, seismic reflections in salt rocks pose a challenge to seismic imaging researchers due to their distinct acoustic characteristics and complex geometry. The boundaries of saline bodies can be identified by trained experts [1], but this process involves massive amounts of seismic data and is labor intensive, making it a potential candidate for being handled by smart image segmentation methods. Nevertheless, public databases of annotated seismic images are scarce, and such data is needed for training smart seismic image segmentation methods based on machine learning schemes, such as RESNET [2] and UNET [3], which are specialized in image segmentation, analysis and recognition.

Therefore, researchers have recently directed their efforts towards data augmentation through seismic image synthesis [4]. Seismic images present unique challenges due to their

non-stationary nature, where texture patterns vary significantly across different geological structures. Salt bodies, in particular, exhibit distinct acoustic properties that generate characteristic boundary patterns in seismic data. Understanding these characteristics is essential for developing effective synthesis methods that are able to create realistic seismic images.

In this work, we propose a mechanism for synthesizing annotated seismic image samples containing salt domes. We use a combination of Variational Autoencoder (VAE) and a context-oriented texture synthesis method to generate new seismic samples containing salt domes. The VAE scheme comprises an encoder that maps the multidimensional input into a low-dimensional latent space, and a decoder that reconstructs the output from these latent features. We wish to minimize the difference between the input and the reconstructed seismic image samples [5]. On the other hand, visual texture synthesis learns to generate new seismic samples from

a particular seismic image by inferring its generation process directly from the seismic image examples provided. The generated textures should be indistinguishable from real data from the human experts point of view. If the experts cannot distinguish between an original texture and a synthesized one, the synthesis process is considered successful [6].

The main research hypothesis of this work is that the combination of VAEs for generating geometric masks with context-oriented texture synthesis can generate synthetic seismic images of salt domes that are virtually indistinguishable from real seismic images by geoscience experts. This hypothesis addresses a critical scientific problem in the field of seismic imaging, analysis and interpretation. This scientific problem arises from the fundamental requirements of modern deep learning models for salt body segmentation, analysis and interpretation. These models demand large volumes of annotated data to achieve robust performance and generalization. However, seismic image annotation is an expensive and time-consuming process that requires specialized expertise and trained geoscientists. The limited availability of annotated datasets significantly restricts the model generalization capabilities, which is particularly concerning given that offshore oil exploration critically depends on the precise identification of salt structures for successful reservoir characterization and drilling operations. This work innovates by combining VAEs to generate salt and rock shapes with a context-oriented texture synthesis process, which handles non-stationary textures of salt bodies, conventional rocks, and their boundaries.

The remainder of this paper is organized as follows. Section II reviews the related work in seismic image synthesis and generative models. Section III presents the proposed methodology, detailing the VAE architecture for context generation, the context-oriented texture synthesis approach, and the complete seismic image synthesis pipeline. Section IV discusses the experimental setup, dataset description, evaluation measures, and presents both qualitative and quantitative results, including a comprehensive comparison with the state-of-the-art method available. Section V presents an ablation study analyzing the contribution of each component of the proposed methodology. Finally, Section VI concludes the paper with a discussion of practical applications and suggests directions for future work.

II. RELATED WORK

OVER the past decades, a variety of algorithms have been proposed to address the challenges posed by seismic image synthesis. In the early days, before the rise of neural networks, researchers relied on traditional methods like model-based approaches and basic texture synthesis techniques. These methods worked reasonably well for simple tasks, but often struggled with complex geological features and the non-uniform textures commonly found in seismic data. The primary limitation of these early techniques was their inability to handle non-stationary textures, which are textures that change across the different regions of a seismic image [7].

In recent years, the introduction of deep learning techniques in seismic image synthesis has advanced significantly the field of seismic image synthesis [1]. The introduction of Convolutional Neural Networks (CNNs) was a major turning point, since it brought a new level of sophistication to seismic image synthesis [8]. CNNs excel at extracting detailed features from images, such as horizontal and vertical lines, and can even synthesize entire objects within a seismic image. The CNNs capabilities allowed for more accurate and realistic synthesized seismic images, making it easier to train methods for the analysis and interpretation of subsurface structures [8].

However, the success of CNNs came with a significant challenge: learning schemes for synthesizing such seismic image structures require large, annotated datasets for training. In seismic imaging, such datasets tend to be unavailable, or difficult to produce. Therefore, researchers have explored various alternatives, including data augmentation with synthetic data and transfer learning. Despite these efforts, the issue of limited training data remains a hurdle in the widespread application of CNNs to seismic image synthesis [11].

More recently, R. S. Ferreira *et al.* [17] proposed a scheme for generating synthetic seismic images. In addition to using expert evaluation, they also evaluated their results using measures that estimate the differences between synthetic and real images. Their work addresses the generation of synthetic seismic images of various geological fields, including geological faults and salt domes. Since the work of R. S. Ferreira *et al.* [17] approaches the same problem and is a clear representative of the state-of-the-art in the field, we compare their results with the results obtained by the proposed scheme using the same measures and methodologies, evaluating the realism of synthetic seismic images when contrasted with real seismic images.

Similarly, Henriques *et al.* [4] proposed a very similar data augmentation method based on training two generative models to augment the number of samples in a seismic image dataset for the semantic segmentation of salt bodies. However, they perform an indirect evaluation of results, based on the improvement in the performance of segmentation methods based on convolutional neural networks, which could be difficult to use as a direct comparison of results with our proposed method and on a different database.

Choi *et al.* [13] address the scarcity of labeled seismic data for training deep learning models by employing advanced generative techniques. They propose a method for generating synthetic seismic images using diffusion models, a class of generative models proficient in various image synthesis tasks. While their primary focus is on generating synthetic seismic data for fault detection, their conditional diffusion model incorporates geological context, ensuring the synthetic images are not only realistic, but also are geologically plausible. Their study demonstrates that diffusion models can effectively capture the complex patterns and textures in seismic imaging data, yielding high-quality synthetic images. Their results confirm that synthetic seismic images enhance the training of fault detection algorithms, leading to an improved

performance in identifying subsurface structures.

In summary, while there has been significant progress in the field of seismic image synthesis, some challenging issues remain open in terms of data availability, evaluation methodologies, and the need for more robust generative models. Our work addresses these challenges by leveraging VAEs and advanced texture synthesis techniques to produce high-quality synthetic seismic images.

III. PROPOSED METHODOLOGY

THE proposed context-Oriented Seismic Image Synthesis methodology combines a deep learning model based on a VAE, with a texture synthesis algorithm. It synthesizes seismic image samples for data augmentation within a specific context mask. The proposed approach consists of two main stages: (1) context generation using a VAE that generates salt body masks corresponding to a given salt geometry, and (2) texture synthesis using a non-parametric algorithm that creates realistic seismic images based on the generated masks. The proposed method focuses on context zones and on the characteristics of seismic images with salt domes, ensuring that the synthetic samples preserve the structural and textural properties of real seismic imaging data. Figure 1 illustrates the complete synthesis pipeline, showing how the VAE-generated masks are combined with context-oriented texture synthesis to produce realistic synthetic seismic images.

A. CONTEXT GENERATION USING A VARIATIONAL AUTOENCODER

The proposed data augmentation scheme detects and generates new rock geometries with a VAE, as illustrated in Fig. 2.

VAEs are deep generative models that learn to encode seismic imaging data into a lower-dimensional latent space, and then decode it back to the original space [9]. VAEs are particularly suitable for generating new seismic imaging samples that follow the same distribution as the training data, making them ideal for data augmentation tasks in seismic imaging.

Fig. 2 illustrates a dataset of seismic images used to generate new salt dome masks. The natural intersection between two different rocks are their boundaries, and in the case of salt rocks, this boundary usually is apparent, as Fig. 3 shows. During the VAE training process, we use only salt masks with a considerable amount of salt, varying from 10 to 90 percent of the total area.

We use the generated masks as contexts for the texture synthesis process. Once we locate the boundaries of a salt body, we assume that its interior region is constituted by salt. The VAE model used in this work employs a simple feed-forward network for the encoder and decoder. The encoder contains four stacked dense layers, and the decoder contains five stacked dense layers. During inference, we sample the latent variables z from the prior distribution $\pi(z)$, and then decode z with the VAE's decoder to generate the salt masks with the inferred distribution.

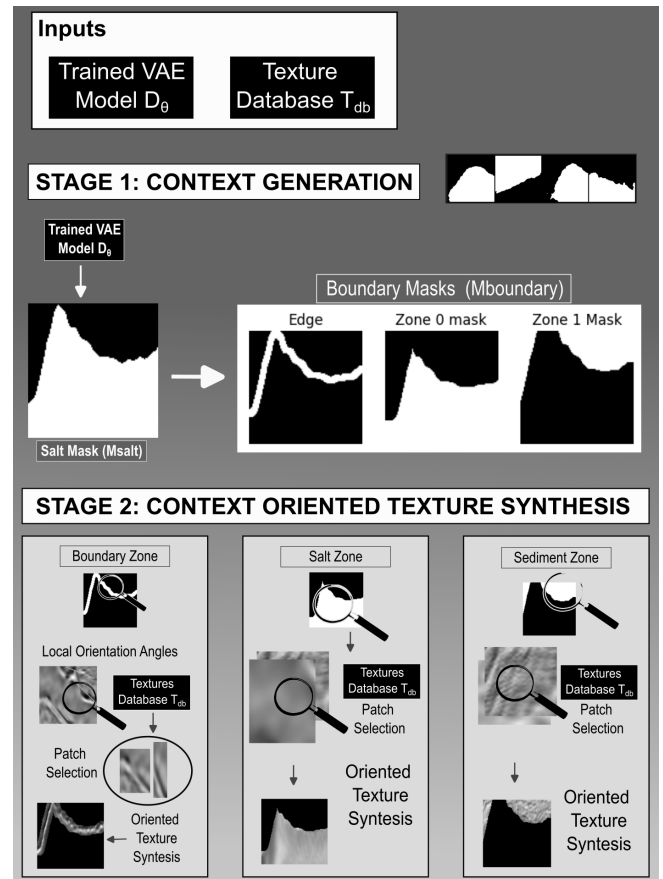


FIGURE 1. Complete pipeline of the proposed context-oriented seismic image synthesis methodology, showing the integration of VAE-based mask generation and the context-aware texture synthesis method.



FIGURE 2. Samples of generated structural salt masks. The Variational Autoencoder learns the features distribution and produces new structural masks.

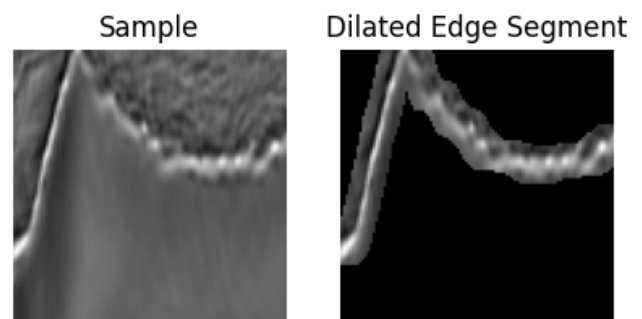


FIGURE 3. Sample of the boundaries identification between salt and other sediments represented in a dilated edge segment.

The main task of VAEs is to model the underlying probability distribution of a data set $x = \{x^{(1)}, x^{(2)}, \dots, x^{(N)}\}$, where each $x^{(i)}$ represents an individual sample. The model introduces a latent variable z (a lower-dimensional representation), and parameters θ to maximize the marginal likelihood $p_\theta(x)$, where x is the observed data:

$$p_\theta(x) = \int p_\theta(x|z)p(z)dz, \quad (1)$$

where $p_\theta(x|z)$ is the likelihood of x given z , and $p(z)$ is the prior distribution over the latent variables.

The key VAE components are the following:

a: Encoder (Recognition Model)

Maps an input seismic image sample x to a latent variable z , and approximates the posterior distribution $q_\phi(z|x)$ (parameterized by ϕ) assuming it to be Gaussian:

$$q_\phi(z|x) = \mathcal{N}(z; \mu_\phi(x), \Sigma_\phi(x)), \quad (2)$$

where $\mu_\phi(x)$ and $\Sigma_\phi(x)$ are the mean and the covariance of the posterior distribution $q_\phi(z|x)$, which are outputs of the encoder network.

b: Decoder (Generative Model)

Reconstructs the data from the latent variable z , modeling the likelihood $p_\theta(x|z)$ (parameterized by θ) assuming it as Gaussian:

$$p_\theta(x|z) = \mathcal{N}(x; \mu_\theta(z), \Sigma_\theta(z)), \quad (3)$$

where $\mu_\theta(z)$ and $\Sigma_\theta(z)$ are the mean and covariance of the likelihood distribution, which are outputs of the decoder network.

c: Latent Variables Priors $p(z)$

Usually assumed as a standard normal distribution:

$$p(z) = \mathcal{N}(z; 0, I), \quad (4)$$

where I is the identity matrix.

d: Variational Inference

Since the exact posterior $p(z|x)$ usually is intractable, VAEs use variational inference to approximate it by optimizing the evidence lower bound (ELBO), where $\text{KL}(\cdot \| \cdot)$ denotes the Kullback-Leibler divergence:

$$\log p_\theta(x) \geq \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)] - \text{KL}(q_\phi(z|x) \| p(z)), \quad (5)$$

and the ELBO has two important components, namely, the reconstruction loss and the Kullback-Leibler (KL) divergence loss, that are detailed next. The reconstruction loss is the expected log-likelihood of the seismic imaging data, which encourages an accurate seismic image sample reconstruction:

$$\mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)], \quad (6)$$

and the KL divergence loss is the Kullback-Leibler divergence between the approximate posterior and the prior, which acts as a regularizer. For Gaussian distributions, this can be computed in closed form as:

$$\text{KL}(q_\phi(z|x) \| p(z)) = \frac{1}{2} \sum_{i=1}^d (\mu_\phi(x)_i^2 + \Sigma_\phi(x)_{ii} - \log \Sigma_\phi(x)_{ii} - 1), \quad (7)$$

where d is the dimensionality of the latent space, and $\mu_\phi(x)_i$ and $\Sigma_\phi(x)_{ii}$ denote the i -th component of the mean vector and the i -th diagonal element of the covariance matrix, respectively.

To enable gradient-based optimization, we use the reparameterization trick. Instead of sampling $z \sim q_\phi(z|x)$ directly, we express z as:

$$z = \mu_\phi(x) + \Sigma_\phi(x)^{1/2} \odot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I), \quad (8)$$

where ε is an auxiliary random variable sampled from a standard normal distribution, and $\odot$ denotes element-wise multiplication. This formulation allows gradients to be propagated through $\mu_\phi(x)$ and $\Sigma_\phi(x)$.

Finally, we train VAEs by maximizing the ELBO with respect to the parameters θ and ϕ using stochastic gradient descent. The optimization process tends to balance the reconstruction accuracy with the latent space regularization.

1) Motivation for Using Variational Autoencoders (VAEs) in Seismic Image Synthesis

The primary reasons for using VAEs are a realistic generation of synthetic samples, an increase in dataset diversity, a regularization and control of variability, and an improvement in the generalization of the segmentation model. When we consider specific reasons related to the nature of seismic data and the purpose of sample generation for data augmentation, VAEs enable the generation of synthetic images that preserve the structural characteristics of real seismic data, thereby enhancing the training and performance of segmentation models [4], [11], [12].

While VAEs offer several advantages for seismic data synthesis, it is important to consider alternative generative models that have also been explored in the literature. In the following sections, we briefly discuss Generative Adversarial Networks (GANs) and Diffusion Models, highlighting their main limitations in the context of seismic image generation and data augmentation.

2) Limitations of Generative Adversarial Networks (GANs) for Seismic Image Synthesis

1) Control over Data Variability:

- It is essential that the generated images preserve the structural patterns of salt bodies. VAEs ensure that synthetic images fall within the expected distribution of real seismic images. GANs, on the other hand, are more prone to generating images

with high variability and may occasionally create unrealistic or inconsistent samples [7].

2) Regularization and Smoothness in Image Generation:

- VAEs impose a probabilistic structure on the latent space, resulting in smoother transitions between different samples and generating images that maintain coherent structural characteristics. GANs, in turn, can suffer from mode collapse [7], where they generate only a limited subset of possible data variations, reducing the diversity of the augmented dataset.

3) Training Stability:

- GANs require a balance between the generator and discriminator, which can make training unstable and more challenging to converge to a distribution that well represents the seismic data [7].

Although GANs are known for generating high-quality images, we chose VAEs in this work due to their more refined control over data variability, training stability, and preservation of the latent seismic data distribution.

3) Limitations of Diffusion Models for Seismic Image Synthesis

Diffusion models present practical and conceptual disadvantages as compared to VAEs [14]:

1) Higher Computational Cost:

- Diffusion models require many sampling steps to generate images, typically hundreds or thousands of denoising steps [14]. This makes them more computationally expensive than VAEs, which generate samples in a single pass through the model.
- In seismic applications, where there is a need to generate a large number of samples for data augmentation, the efficiency of VAEs can be a significant advantage.

2) Need for Large Volume of Training Data:

- Diffusion models often require large training datasets to capture high-quality distributions [14]. Since labeled seismic images can be limited, VAEs may be more effective in learning useful representations even with a smaller number of samples.

Therefore, targeting efficiency and ease of training, VAEs have simpler and more stable training, as they do not rely on an iterative refinement process like diffusion models. This can be a decisive factor when there are time and computational resource constraints.

B. CONTEXT-ORIENTED TEXTURE SYNTHESIS

Texture synthesis is a technique used in computer graphics and image processing to generate new textures that are visually similar to a given sample texture. Non-parametric texture synthesis methods [10] create new images by sampling and recombining patches from sample textures provided as examples, ensuring that the statistical properties of the generated

texture closely match those of the texture samples provided as the original examples.

The typical texture synthesis algorithms use an image database and replicate the images in a picture plane according to some arrangement rules. Also, texture synthesis can be enhanced by working at a finer scale, from the pixel level to a small region level. At each iteration of the algorithm, a small portion of the original texture, a patch, is copied and placed within the newly synthesized texture. Non-parametric sampling [10] further improved texture synthesis by creating a new image that shares the same statistical properties as a given sample texture, while avoiding explicit models of the texture by sampling directly from the example texture images.

1) Non-parametric Seismic Texture Synthesis Algorithm

In order to detail the proposed seismic texture synthesis method, the following elements must be defined :

1. Neighborhood: For each pixel p in the image to be synthesized, a region around p already filled with known texture pixel values is defined (namely, a neighborhood $N(p)$);

2. Similarity Criterion: To find corresponding patches in the example texture image, we compute the similarity between $N(p)$ and the neighborhoods $N(q)$ for all pixels q in the example texture image. We define as a similarity measure the Euclidean distance D between the pixels of the neighborhoods:

$$D(N(p), N(q)) = \sum_{i \in N} (I_p(i) - I_q(i))^2, \quad (9)$$

where $I_p(i)$ and $I_q(i)$ are the pixel intensity values at location i in the neighborhoods of pixels p and q , respectively, and N denotes the set of pixel locations in the neighborhood $N(p)$ of a pixel p ;

3. Random Sampling: We choose a pixel q randomly from the example texture image with probability proportional to its similarity (or inverse of $D(N(p), N(q))$) to p . In other words, pixels with smaller distance $D(N(p), N(q))$ are more likely to be chosen. We can model the probability $P(q)$ of selecting pixel q as:

$$P(q) = \frac{\exp(-D(N(p), N(q))/\sigma^2)}{\sum_{q'} \exp(-D(N(p), N(q'))/\sigma^2)}, \quad (10)$$

where σ is a temperature parameter that controls the influence of distances in the pixel selection probability, and q' denotes all candidate pixels in the example texture image;

4. Image Filling: The process begins with a small initial patch in the new image and gradually expands it, pixel by pixel, using the steps described above to fill the entire texture image that is being synthesized. The choice of the next pixel to be synthesized follows a raster or spiral filling path. The method produces seismic textures that are visually similar to the original input texture.

2) Context-based Seismic Image Synthesis

We constrain the process of seismic image synthesis to the context masks generated by the VAE model. Various patches

are used as inputs, corresponding to different parts of the sample seismic image. The synthesized sample has three main components, namely, salt zone, boundary zone, and conventional rock zone. Initially, we create the boundary, or frontier, zones between the two rocks, which we also call edge zone. We create the boundary zone by using edge detection on the original seismic image to locate the interfaces between the two types of rocks, and then make it thicker, creating an edge strip. As Fig. 3 shows, this strip is an area of high seismic contrast appearing as light and dark bands. Also, various parallel lines make up this boundary, and Fig. 4 shows their angles. We use a dataset composed of edge pieces and their corresponding angles as input for the boundary synthesis process. Then, for each edge segment in the mask of the seismic texture image to be synthesized, we synthesize its texture based on the patch with the most similar angle (see Fig. 5). We repeat this process until all the segments of the edge band are covered.

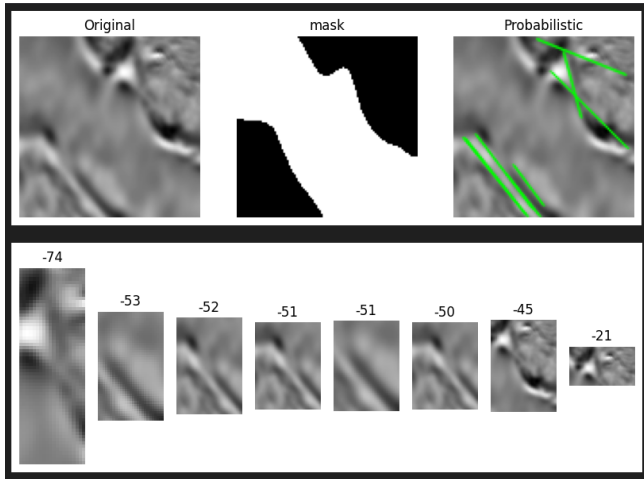


FIGURE 4. The green lines at original image identify border zones and angles to guide the patch selection in the patch database construction process.

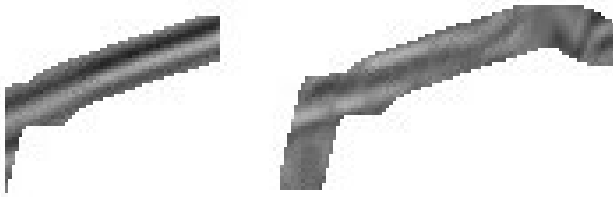


FIGURE 5. The Construction of new boundaries in the generated seismic images uses texture synthesis based on selected samples from the seismic image patches database.

To synthesize the salt zones, we use the corresponding type of image region as a seismic texture reference to generate the new synthesized pixels. Finally, we synthesize the conventional rock zones using the corresponding sediment regions in the seismic texture sample provided as an example. We divide the seismic image synthesis into steps to conform to the specific characteristics of this kind of seismic image, such as

the edge, saline rock, and conventional rock zones, as Fig. 6 shows.

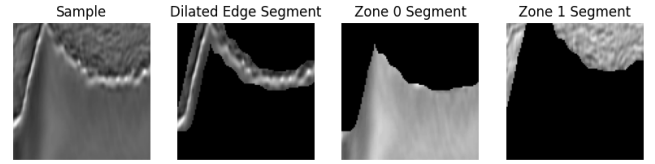


FIGURE 6. Real seismic saline images divided by context zones used as input to the texture synthesis process.

C. COMPLETE SEISMIC IMAGE SYNTHESIS PIPELINE

The complete synthesis pipeline integrates the VAE-based context generation and the context-oriented texture synthesis methods. The proposed seismic image synthesis algorithm processes all image zones, one by one, until the seismic image synthesis process is completed, according to their corresponding zone samples in the input seismic image data. The pipeline can be summarized as follows: (1) the VAE method generates a new salt body mask defining the geometric structure of the synthesized seismic image, (2) edge detection is used to identify boundary zones between salt and sediment, (3) the texture synthesis algorithm fills each zone (edges, salt, and sediment) with the appropriate seismic textures based on the training seismic image database, and finally (4) the final synthetic seismic image is assembled by combining all synthesized zones. This systematic approach, detailed in Algorithm 1, which was designed to ensure that the synthesized seismic images maintain both structural consistency and textural realism.

IV. RESULTS AND DISCUSSION

THIS section presents and discusses the obtained experimental results in order to illustrate the effectiveness of the proposed approach. The experimental evaluation encompasses a qualitative assessment performed by domain experts, and a quantitative comparison with the available state-of-the-art. This dual evaluation strategy follows the criteria below :

- 1) **Qualitative Evaluation by Experts — Blind Discrimination Experiment:** Experts try to identify the portions of salt dome structures inside the seismic samples, and each expert produces a mask label that will be compared with the mask generated by our proposed method. To rigorously assess the perceptual realism of the synthetic images, a blind discrimination experiment is being conducted. In this protocol, real and synthetic seismic images are presented in a randomized, interleaved order, and expert geoscientists are asked to classify each image as real or synthetic, without prior knowledge of its origin. This design constitutes a formal blind discrimination test, providing stronger evidence for the “virtually indistinguishable” hypothesis than a sequential evaluation. Quantitative

Algorithm 1 Context-Oriented Seismic Image Synthesis

```

1: Input: Trained VAE model  $D_\theta$ , Texture Database  $T_{db}$ 
2: Output: Synthetic Seismic Image  $I_{synth}$ 

// Stage 1: Context Generation
3: Sample latent vector  $z \sim \mathcal{N}(0, I)$ 
4: Generate salt mask  $M_{salt} \leftarrow D_\theta(z)$ 
5: Identify boundary mask  $M_{boundary}$  from  $M_{salt}$  via edge
   detection and dilation
6: Define sediment mask  $M_{sediment} \leftarrow \neg(M_{salt} \cup M_{boundary})$ 
7: Initialize  $I_{synth}$  as an empty image

// Stage 2: Context-Oriented Texture Synthesis
8: // Synthesize Boundary Zone
9: for each segment  $s$  in  $M_{boundary}$  do
10:   Compute local orientation  $\alpha_s$ 
11:   Select best patch  $P_{boundary} \in T_{db}$  based on  $\alpha_s$ 
12:   Synthesize texture for  $s$  in  $I_{synth}$  using  $P_{boundary}$ 
13: end for
14: // Synthesize Salt Zone
15: Select salt patches  $P_{salt} \in T_{db}$ 
16: Synthesize texture for region  $M_{salt}$  in  $I_{synth}$  using  $P_{salt}$ 
17: // Synthesize Sediment Zone
18: Select sediment patches  $P_{sediment} \in T_{db}$ 
19: Synthesize texture for region  $M_{sediment}$  in  $I_{synth}$  using
    $P_{sediment}$ 
20: return  $I_{synth}$ 

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measures –such as the F1 score –are also used to evaluate the difference between the experts' predictions and the proposed method synthesized masks. experts' segmentation accuracy on both real and synthetic images. [TODO: Replace this description with the results of the blind experiment when available.]

- 2) **Quantitative Comparison with the State-of-the-Art:** The quantitative comparison is performed with the method proposed by Ferreira *et al.* [17] since it is representative of the state of the art, and addresses the same problem being tackled in our work. To enable a direct comparison, we reproduced the synthesis experiment using the F3 seismic dataset employed by Ferreira *et al.* [17]. This quantitative comparison is based on three measures: mean squared error (MSE), structural similarity index (SSIM), and the Euclidean distance, using Local Binary Patterns (LBP) for texture attribute assessment. These measures quantify the similarity between the original seismic images and the synthetic samples generated by the chosen synthesis method. In order to facilitate a direct and fair comparison between methods, we compare our proposed seismic image synthesis approach with the scheme proposed by Ferreira *et al.* [17] using the same measures and seismic dataset.

A. DATASET

The results of the proposed seismic image synthesis scheme are evaluated by experts, and the obtained synthetic image quality is compared with the quality of the results obtained by a comparable seismic image synthesis method. Therefore, two seismic image datasets were used in this work for these different evaluation purposes, which are described below :

a: TGS Salt Identification Challenge Dataset

The salt body dataset from the TGS Salt Identification Challenge is employed for qualitative expert evaluation. It comprises 2D image slices representing a 3D view of the interior of the Earth obtained through reflection seismology. The data is a set of images chosen randomly in the subsurface, with dimensions of 101×101 pixels, where each pixel is classified as either salt or sediment [15]. This dataset is used for the assessment of geoscience domain experts to evaluate the visual quality and geological realism of the synthetic seismic images obtained by our proposed scheme.

b: F3 Seismic Dataset

The F3 dataset, containing 3D post-stack seismic data from the Dutch North Sea, is used for a quantitative comparison of the quality of our synthetic seismic images with the image quality obtained by a comparable method representing the state-of-the-art proposed by Ferreira *et al.* [17]. This dataset is widely recognized as a benchmark in seismic interpretation and image synthesis research [16]. Using the same dataset and evaluation protocol as the method in [17] ensures a direct and fair comparison based on quantitative measures (MSE, DSSIM, and LBP Distance).

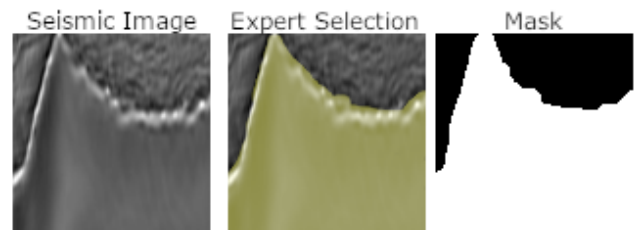


FIGURE 7. Comparison between a real seismic image sample, and the geoscience specialist selection in terms of saline area and ground truth (saline mask).

B. EVALUATION MEASURES

Through qualitative analysis, the geoscience specialist evaluates the seismic image samples to identify the portions of salt bodies in these images. To assess the quality of the seismic image synthesis, the precision, recall, and F1-score measures are used to compare the synthetic seismic image with the ground truth, as detailed below:

The F1 score is a widely used measure for evaluating binary classification models. It is particularly useful when a balance between precision and recall is desired. It allows access to the quality of the classification predictions. While the correct classification rate is simple to understand, it can be

misleading in imbalanced datasets. Therefore, measures such as precision, recall, and the F1 score tend to provide a clearer scenario [8].

- **Precision:** is the proportion of true positives (TP) over all predicted positive results (TP + FP)

$$\text{Precision} = \frac{TP}{TP + FP} \quad (11)$$

- **Recall:** or sensitivity, is the proportion of true positives over all actual positive cases (TP + FN)

$$\text{Recall} = \frac{TP}{TP + FN} \quad (12)$$

- **F1 Score:** is the harmonic mean of precision and recall, providing a single measure that considers both.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (13)$$

The F1 score is particularly valuable in classification situations, where the positive class tends to be rare. It helps to understand the trade-off between false positives and false negatives. Usually, the F1 score provides a balanced measure that is less susceptible to class distribution bias.

We use the precision, recall, and F1-score measures to evaluate the salt bodies masks generated by the experts, and we can compute them using the confusion matrix derived from these expert-generated masks. To evaluate the quality of the synthesized samples, we evaluate the overlap between the experts selections of pixels corresponding to the salt bodies in synthetic images and synthetic images masks, as illustrated in Fig. 7. Fig. 9 illustrates the ground truth image and the specialist evaluation of the salt bodies locations, and the computed precision, recall, and F1-score measures for this particular case.

The quantitative comparison followed the methodology proposed by Ferreira *et al.* [17], where each synthetic image is compared with the original seismic image that was used as its template. To this purpose, and also following the methodology in [17], we use the measures mean squared error (MSE, Eq. 14), Structural Similarity Index (SSIM, represented as DSSIM in Eq. 15), and Euclidean distance between the original template image and the synthetic images based on the Local Binary Pattern (LBP) texture attribute.

The MSE measure defines a pixel-to-pixel distance between two images. We calculate the MSE between two images by subtracting each synthetic pixel value from the original (real) image, square the pixel difference as the error (to eliminate negative signs and penalize larger errors), and we finally take the average of these errors squared :

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad (14)$$

where n is the total number of pixels, y_i is the i -th pixel value in the original image, and $\hat{y}_i$ is the corresponding pixel value in the synthetic image.

The SSIM measures the perceptual similarity between two images x and y by incorporating texture information, such as

variances and covariances, which is widely used in computer vision and image processing to assess the quality of image reconstructions, compressions, or transmissions. It evaluates luminance (brightness), contrast, and structural attributes as local patterns and shapes. Since SSIM values are defined in the range $[-1, 1]$, the common transformation below represents it as a distance to facilitate comparison, and from now on, it is called DSSIM:

$$\text{DSSIM}(x, y) = \frac{1 - \text{SSIM}(x, y)}{2} \quad (15)$$

The interpretation of DSSIM values as a distance between 0 and 1 is as follows:

- DSSIM = 0: the images are identical;
- DSSIM from 0.005 to 0.10: high similarity;
- DSSIM > 0.25: low perceptual similarity;
- DSSIM from 0.5 to 1: very different.

The LBP is a more robust texture attribute successfully applied in medical image analysis and in geosciences [18], [19]. The LBP calculation uses four neighbors, distance 1, and a 64-pixel tile size. The calculation produces a histogram for each image, representing the frequency of local binary patterns. Finally, we compute the Euclidean Distance between the histograms. Therefore, the smaller the distance, the more similar the textures of the two images are. Otherwise, larger histogram distances represent different patterns.

We conducted the qualitative evaluation of the synthesized seismic image samples through a systematic process with the support of geoscientists, who are experts in the field. Our goal is to evaluate the synthesized samples quality, relevance, and reliability using the knowledge and experience of qualified professionals. Three geoscientists who routinely work in seismic analysis and interpretation participated in this evaluation. We conducted the evaluation by having experts reviewing the seismic images, and selecting different regions types by painting the salt bodies. Images is being redesigned as a formal blind discrimination experiment, following the protocol described below.

1) Blind Discrimination Experiment Protocol

In the redesigned evaluation, real and synthetic seismic images from the TGS Salt Identification Challenge dataset are presented to expert geoscientists in a single, randomized, interleaved sequence. Experts are *not* informed of the origin of each image prior to evaluation. For each image, the experts are asked to:

- 1) Classify the image as *real* or *synthetic*;
- 2) Identify and segment (mark) the salt body regions using a graphics computer program.

The segmentation masks produced by the experts are compared against the ground-truth masks using the precision, recall, and F1-score measures described in Section IV-B were calculated for each expert and for the group of experts, and we consolidate their averages and standard deviations. Table ?? shows

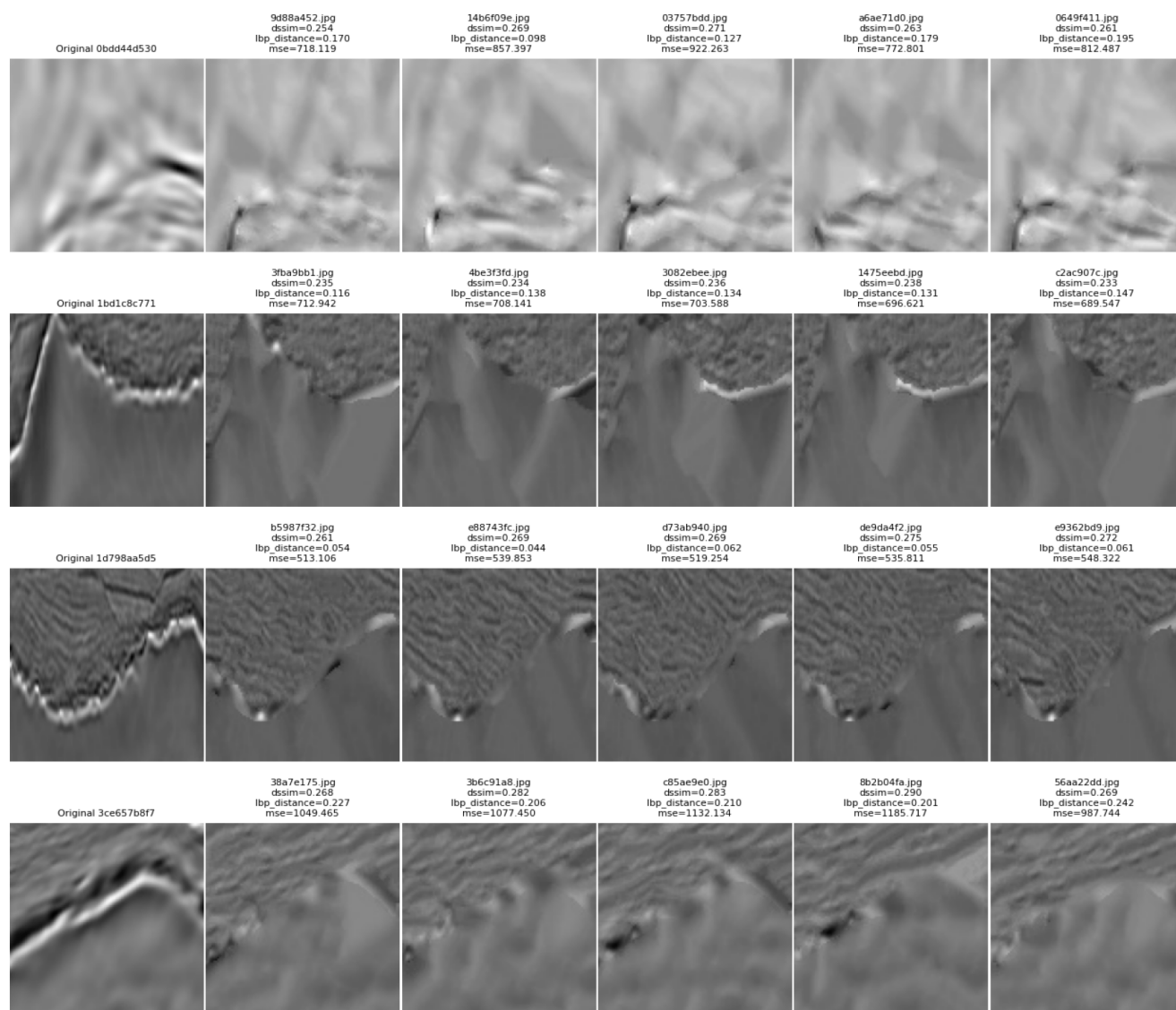


FIGURE 8. Synthetic Seismic Samples.

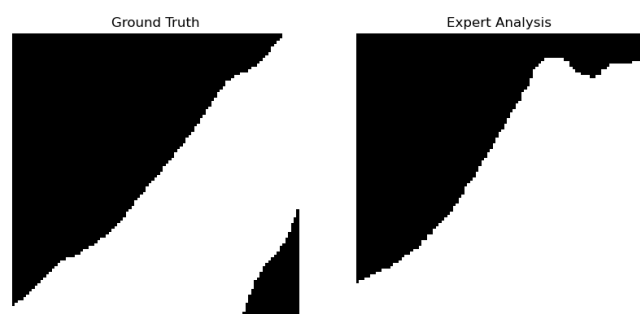


FIGURE 9. Quantitative analysis illustration using sample measurements for comparing the ground truth and expert analysis: Precision: 0.88 ; Recall: 0.87 ; F1-score: 0.87.

these evaluations for 20 real seismic images, showing an average precision score of 0.88761, suggesting that experts identifications of saline rock portions were 88.7% accurate, leaving 11.3% misidentified as salt, which was actually common rock. The obtained average recall of 0.87795 indicates that 87.7% of the saline rock portions were effectively identified, with a small remainder of 12.3% missed salt portions. Considering the F1-score calculation, the specialists achieved an average score of 0.88159 for the . This design constitutes a formal blind discrimination test, providing evidence directly relevant to the hypothesis that the synthetic images are virtually indistinguishable from real seismic images by geoscience domain experts.

The evaluation involves three geoscientists who rou-

tinely work in seismic analysis and interpretation. Control images without any saline body are included in the stimulus set to assess the experts' baseline response tendency. The identity and sequence of images are fully randomized across participants to avoid order effects.

Comparing the results of the geoscience specialist evaluation of real and synthetic seismic images, a small difference of less than 2% can be observed in favor of the real images. Therefore, from the perspective of the specialists evaluations, the synthetic seismic images are virtually indistinguishable from the real seismic images. In that sense, we can conclude that, from the point of view of the geoscience specialists consulted, the synthetic images are comparable to the real seismic images.

TODO: The blind discrimination experiment is currently being conducted. The results — including expert classification accuracy (real vs. synthetic), and segmentation F1-scores for both real and synthetic images — will replace this placeholder in the revised manuscript. Tables analogous to those reported in the previous sequential evaluation will be provided here.

C. QUANTITATIVE EVALUATION

In order to make a direct and fair comparison with the results obtained using the approach proposed by Ferreira *et al.* [17], we repeated their synthesis experiment using the F3 seismic dataset employed in their original work. The F3 dataset contains 3D post-stack seismic data from the Dutch North Sea, widely used as a benchmark in seismic interpretation and image synthesis [16]. Following the same evaluation protocol, each synthetic image is compared against the original seismic image that served as its template, using MSE, DSSIM, and LBP Distance as described in Section IV-B. The quantitative results obtained by our proposed method on the F3 dataset are presented in Table 1 and discussed in the comparison below.

D. COMPARISON WITH THE STATE-OF-THE-ART

In order to provide a quantitative comparison based on the F3 dataset, we evaluated 600 generated synthetic images created by our proposed method and compared them with the sketch-based approach proposed by Ferreira *et al.* [17]. Our results are shown in Table 1 alongside the results reported by Ferreira *et al.* [17] in their work. Both methodologies measure the similarity between synthetic and original seismic images using three key measures: Mean Squared Error (MSE), Structural Similarity Dissimilarity (DSSIM), and Euclidean Distance based on Local Binary Patterns (LBP Distance). For these measures, lower values signify better performance.

The method proposed by Ferreira *et al.* [17] employs a conditional Generative Adversarial Network (cGAN) architecture based on the pix2pix framework to synthesize seismic images from original (real) seismic images with user-provided sketches. The generator network takes a sketch as

input and learns to produce realistic seismic images matching its structural layout, while a discriminator evaluates the realism and alignment of the generated images. Their approach requires a step of manual sketch creation as a conditional input, whereas our proposed method based on context-oriented texture synthesis operates autonomously, generating the corresponding seismic image textures without requiring human intervention.

1) Comparative Analysis

The following subsections present a detailed comparison based on the median values of the measurements obtained for the F3 dataset, since these median values provide an estimate of the typical performance of compared methods. The definitions of MSE, DSSIM, and LBP Distance are provided in Section IV-B.

The sketch-based method [17] achieved a median MSE of 4712.1 using its best-performing configuration. Our proposed context-oriented approach achieved a median MSE of 3926.2325, potentially providing a substantial improvement in terms of pixel-wise correspondence between the synthetic and original images.

In Ferreira *et al.* [17], the median DSSIM achieved 0.39 using its best-performing configuration. Our proposed context-oriented approach achieved a median DSSIM of 0.389125. While the improvement in terms of DSSIM is modest, it still suggests a better structural similarity between the synthetic and original images as compared to the approach in [17]. The relatively small difference in DSSIM suggests that both methods are capable of capturing the overall structure of the seismic images, but our proposed method may have an advantage in terms of providing finer details.

The work in Ferreira *et al.* [17] reported the lowest median LBP Distance value of 0.17. In contrast, our proposed method achieved a LBP Distance median of 0.150075, indicating that our method tends to generate textures with higher fidelity to the original images.

Overall, our proposed context-oriented synthesis approach tends to obtain a better quantitative performance across all three measures when compared to the sketch-based method proposed in Ferreira *et al.* [17]. The improvement is most pronounced in terms of the MSE and LBP distance measures, suggesting notable gains in terms of pixel-level accuracy and texture fidelity.

It is important to observe that quantitative measures may not always align with qualitative human assessment. In our work, we supplemented our quantitative findings with a detailed qualitative analysis by domain experts. The experts evaluations concluded that the synthetic images were virtually indistinguishable from the real world ones, and these findings were supported by an F1-score difference of less than 2%.

TABLE 1. Quantitative comparison based on the F3 dataset (the lower the better).

Metric	Ferreira et al.	Proposed
MSE		
Min	3720.4	1255.53
Median	4712.1	3926.23
Max	5931.6	6686.15
DSSIM		
Min	0.3381	0.3482
Median	0.3978	0.3891
Max	0.4528	0.4240
LBP Distance		
Min	0.1222	0.1187
Median	0.1714	0.1500
Max	0.2428	0.2050

V. ABLATION STUDY

TO assess the contribution of each component of our proposed methodology, we conducted an ablation study, discussed below. We design these experiments to isolate the impact of the VAE-based context generation and the context-oriented texture synthesis in the final quality of the synthetic seismic images.

A. IMPACT OF CONTEXT-ORIENTED SYNTHESIS

A key innovation of our method is the specialization of the synthesis process in three distinct zones: salt, conventional rock, and their boundary. To evaluate the importance of this specialization, we replace the zone-specific approach with a single nonparametric texture synthesis model that generates the entire image guided only by the binary salt/non-salt mask, without discriminating between salt dome interior, surrounding rock, and boundary regions. Removing this context-oriented step tends to reduce the zone-specific textural fidelity, preventing an accurate correspondence with annotated masks, causing a worse geological interface reproduction. The experimental results are compared with the full context-oriented synthesis method using MSE, DSSIM, and LBP Distance.

Results and Analysis: We conducted the ablation experiment by synthesizing 110 images using a single unified texture synthesis model without zone-specific processing. Critically, 44 images (40%) had to be discarded due to lack of geological coherence, demonstrating the fundamental limitation of non-contextualized synthesis. For the remaining valid samples, the quantitative evaluation revealed median values of MSE = 837.38, DSSIM = 0.313, and LBP Distance = 0.094. In comparison, our context-oriented method consistently achieves superior performance with typical median values of MSE < 800 (best group: 542.87), DSSIM < 0.30 (best group: 0.2424), and LBP Distance < 0.09 (best group: 0.0800). These results confirm that even after filtering out geologically incoherent samples, the ablation approach exhibits poorer performance across all metrics. The higher MSE indicates reduced pixel-level fidelity to the original images, the increased DSSIM reflects lower structural similarity,

and the elevated LBP Distance demonstrates inferior texture preservation. The necessity to discard 40% of the ablation samples, combined with the quantitatively inferior quality of the remaining images, tends to validate that the full context-oriented approach is required for generating high-fidelity seismic images that maintain accurate correspondence with the annotated geological zones.

B. CONTRIBUTION OF THE VAE FOR MASK GENERATION

Our method relies on a VAE to generate realistic and diverse salt body geometries. To quantify the benefit of using a VAE, we generated the following two sets of 200 masks: (1) VAE samples drawn from the latent space, and (2) geometric transformations (rotation, scaling, horizontal/vertical flipping, elastic deformation) applied to existing ground-truth masks. Because the VAE learns a continuous latent representation of salt geometries, it should produce masks with greater diversity and more realistic geometric characteristics than simple geometric transformations, which are constrained to variations of existing shapes. We compared both sets using pairwise IoU for shape diversity, two-sample Kolmogorov-Smirnov (KS) tests on six shape descriptors (area, perimeter, compactness, eccentricity, solidity, and extent) for geometric realism, and verified the expert geological plausibility ratings on a 5-point scale.

Results and Analysis: The results for both sets against the ground truth (GT, 1,617 masks) are summarized in Table 2. The VAE achieves higher shape diversity (score = 0.765, mean IoU = 0.235) than geometric transforms (0.755, IoU = 0.245) and GT (0.713), confirming that latent-space sampling explores a broader range of salt geometries. In KS tests ($\alpha = 0.05$), the VAE masks are statistically indistinguishable from the GT in 4 out of 6 descriptors (area, perimeter, compactness, solidity), while geometric transforms are statistically indistinguishable from the GT according to 2 descriptors only (compactness, extent). The VAE mean coverage (39.2%) is also closer to the GT (45.6%) than transforms (33.6%). The expert reviewers rated VAE masks higher (3.85/5 vs. 3.48/5), while the geometric-transform samples frequently received the lowest score (1/5) due to their geologically implausible shapes produced by unconstrained operations, such as extreme scaling or flipping—artifacts that the VAE latent space implicitly avoids. These results confirm that the VAE outperforms geometric augmentation across all evaluation dimensions, validating its role as the preferred mask generation component in our proposed approach.

These ablation studies systematically validate our design choices, and provide a deeper understanding of why the proposed context-oriented approach is effective for synthesizing high-fidelity seismic images.

VI. CONCLUDING REMARKS AND FUTURE DIRECTIONS

THIS work introduced a novel context-oriented data augmentation methodology for synthesizing labeled seismic images containing salt dome structures, addressing a critical challenge in offshore petroleum exploration where annotated

TABLE 2. Ablation of Mask Generation: VAE vs. Geometric Transforms (200 masks each; GT = 1,617 masks).

Metric	Ground Truth	VAE Gen.	Geom. Trans.
No. of masks	1,617	200	200
Diversity score	0.713	0.765	0.755
Mean pairwise IoU	0.287	0.235	0.245
Coverage (%)	45.6	39.2	33.6
Area (mean px ²)	4,455	3,820	3,054
Perimeter (mean px)	303.4	289.5	249.2
Compactness (mean)	0.565	0.519	0.524
Eccentricity (mean)	0.827	0.850	0.808
Solidity (mean)	0.909	0.882	0.889
Extent (mean)	0.703	0.625	0.673
KS similar to GT (of 6)	—	4	2
Expert rating (/5)	—	3.85	3.48

training data is scarce and expensive to obtain. By integrating Variational Autoencoders for geometric mask generation with context-aware non-parametric texture synthesis, the proposed approach successfully generates synthetic seismic images that preserve both structural integrity and textural fidelity across three distinct geological zones: salt bodies, conventional rocks, and their boundaries.

The comprehensive evaluation framework, combining expert qualitative assessment with rigorous quantitative measures, validates the effectiveness of our methodology. Three geoscience experts evaluated the synthetic samples, achieving F1-scores that differed by less than 2% from their evaluations of real images (0.86901 vs. 0.88159), demonstrating that the synthetic images are virtually indistinguishable from authentic seismic data. A formal blind discrimination experiment — where real and synthetic images are presented in randomized, interleaved order to expert geoscientists — is currently being conducted to rigorously assess the perceptual realism of the synthesized images. [TODO: Update this paragraph with the results of the blind expert evaluation experiment, replacing the preliminary sequential evaluation results.] Quantitatively, our proposed method tends to outperform the state-of-the-art sketch-based GAN approach on the F3 dataset, considering all three measures proposed in [17] (MSE, DSSIM, and LBP Distance).

The significance of this work extends beyond technical measures. In the petroleum industry, where seismic interpretation requires extensive expertise and labeled datasets are prohibitively expensive, our automated synthesis method offers a practical solution for generating high-quality training data at scale. The context-oriented approach ensures geological plausibility by respecting the distinct characteristics of different rock types and their interfaces, a crucial requirement for real-world applications in subsurface exploration and reservoir characterization.

A. APPLICATIONS

The proposed methodology enables several important applications in the geoscience domain. The synthetic samples can augment limited real datasets to improve the robustness and

generalization of deep learning models for salt body segmentation, such as U-Net and ResNet architectures, particularly where obtaining additional labeled data is impractical. This synthetic data can also serve as a pre-training corpus for transfer learning, allowing models trained on one geological basin to adapt more effectively to different regions with limited labeled samples.

Furthermore, the VAE's ability to interpolate in latent space enables the generation of salt geometries that are underrepresented in training datasets, helping to address class imbalance and improve model performance on edge cases. Beyond model training, these high-quality synthetic seismic images can support educational programs for geoscience professionals, providing diverse examples for teaching seismic interpretation and salt body identification skills. Finally, by generating multiple plausible interpretations of ambiguous seismic regions, the method can support uncertainty analysis in exploration decision-making processes.

B. FUTURE RESEARCH DIRECTIONS

While this work demonstrates promising results on the TGS Salt Identification Challenge dataset, several avenues for future research merit exploration:

Adapting the methodology to synthesize other complex geological features such as faults, channels, and stratigraphic layers would broaden its applicability across different exploration contexts through **Extended Geological Structures**. Building upon this foundation, developing hierarchical **Multi-scale Synthesis** approaches that can generate seismic volumes at multiple resolutions could better capture the multi-scale nature of geological structures and improve computational efficiency.

To enhance the method's generalizability and practical utility, investigating **Domain Adaptation** techniques to adapt the synthesis process across different seismic acquisition parameters, geological basins, and imaging conditions represents a crucial direction. Furthermore, incorporating additional conditioning variables such as depth, geological age, or depositional environment through **Conditional Generation** could provide finer control over the synthesis process and generate more geologically consistent samples.

From a practical deployment perspective, conducting systematic studies on how synthetic data augmentation affects the performance of various seismic interpretation tasks through **Downstream Task Evaluation**—including not only segmentation but also classification, detection, and attribute prediction—would validate the method's broader impact. Additionally, optimizing the computational efficiency of the pipeline to enable **Real-time Synthesis** during active learning workflows, where models request specific types of synthetic samples to address their current weaknesses, would significantly enhance its practical value.

Looking forward, integrating seismic wave propagation physics and rock property constraints into the generation process through **Physics-informed Constraints** could further enhance the geological realism and interpretability of

synthetic samples, representing a promising direction that bridges data-driven and physics-based approaches.

In conclusion, this work establishes a robust foundation for context-oriented seismic image synthesis, demonstrating that carefully designed generative approaches can produce training data of sufficient quality to support advanced machine learning applications in geoscience. The combination of automated mask generation, zone-specific texture synthesis, and rigorous evaluation provides a template for developing similar methodologies in other domains where labeled data scarcity constrains the application of deep learning techniques.

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